IILM UNIVERSITY Greater Noida

Department of Computer Science and Engineering

Introduction to AI & ML Project

Project Title : AI Adaptive Assistance (Athena)

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ABSTRACT

The biggest challenge facing digital literacy in rural India today is fear and a lack of accessible tools. Most training materials are in English, require typing skills that beginners do not have, or create a fear of "breaking" the computer. To solve this, we built **Athena**, an AI-powered teaching assistant designed specifically for first-time users. Unlike standard assistants like Siri or Alexa which just give answers, Athena is a "do-and-show" teacher. It accepts voice commands in the user's local language (removing the need to type), explains the task visually, and then actually performs the operation on the screen—like creating a folder or opening a website—so the user can watch and learn. We also built a "risk-free" training mode where users can practice on demo files without touching personal data. This project proves that by using specific APIs for voice and automation, we can build a tool that understands the learner's pace, making digital education accessible, secure, and personalized for everyone.

INTRODUCTION

1.1 Project Background & Context

In today's world, knowing how to use a computer is as important as reading and writing. However, a massive "digital divide" still exists in our country While urban areas adapt quickly, rural populations are often left behind because they lack trained teachers and simple learning resources. Existing tools are often too technical, predominantly in English, and don't offer the hands-on guidance a beginner needs. This project, Athena, was born out of the need to bridge this gap using Artificial Intelligence. We wanted to create a companion that sits on the desktop and guides a user just like a human teacher would.

1.2 Problem Formulation

The core problem we are addressing is the "fear of technology" and the language barrier.

- ***Language Barrier***: Most digital interfaces are in English, which many rural users struggle with.

- ***Lack of Confidence***: Beginners are often afraid they will delete something important or "mess up" the settings, which stops them from trying.

- ***Lack of Resources***: There aren't enough human teachers to go to every village and teach these basic skills one-on-one.

1.3 Project Objectives

The main goals of the Athena project are:

1. ***Voice-First Interaction***: To allow users to control the computer entirely through voice commands, removing the struggle of typing.
2. ***Visual Demonstration***: To not just tell the user what to do, but to show them by automatically moving the mouse and opening windows[9].
3. ***Risk-Free Learning***: To provide a safe "sandbox" environment where users can practice deleting and moving files without real consequences.
4. ***API Integration***: To successfully combine different technologies—Speech-to-Text, OS automation, and Generative AI—into one smooth workflow.

1.4 Scope and Limitations

- ***Scope***: The project currently focuses on basic operating system tasks (creating folders, opening calculators, web browsing) and general knowledge questions.

- ***Limitations***: Currently, the system relies on an active internet connection for the best voice recognition results (Google/OpenAI APIs). Also, the system is optimized for English voice commands right now, with local language translation being a key part of the future roadmap.

THEORETICAL BACKGROUND & LITERATURE REVIEW

2.1 Principles of Voice-Activated Automation

This project relies on Natural Language Understanding (NLU). Unlike a simple "if-else" code, an AI assistant needs to understand *intent*. When a user says "Make a new folder" or "Create a directory," the system must recognize that both mean the same thing. We utilize specific Python libraries to bridge the gap between spoken words and operating system actions.

2.2 Review of Related Work (2021-2025)

We reviewed several research papers to understand what has already been done in this field.

Personal Assistants for Daily Tasks

- **Mishra, Dhiman, & Dhyani (2024)** developed an "AI Assistant for Daily Use." Their work focused on basic automation but lacked the educational aspect. They showed that Python is efficient for these tasks but didn't address the "teaching" element we are focusing on.
- **Kale et al. (2025)** worked on "EVE – Your Personal AI Assistant." They emphasized responsiveness and threading (doing two things at once). This helped us design Athena so it doesn't freeze while listening to the user.

Desktop Automation

- **Bharadwaj et al. (2024)** explored "Automating Desktop Tasks with Voice." Their research validated our use of libraries like py auto Gui and us to control the mouse and keyboard programmatically. However, their system was purely for execution, not for literacy or training.

AI in Education+

- **Jacob et al. (2024)** discussed AI-based virtual assistants in a broader context. They highlighted how AI can personalize learning, which supports our goal of adapting Athena to the learner's speed.

2.3 Identification of Research Gap

Most existing assistants (like Siri or the ones in the papers above) are designed for **convenience**—doing a task quickly for someone who already knows how to use a computer. There is a gap in tools designed for **education**—teaching someone *how* to use the computer while doing the task. Athena fills this gap by acting as a tutor first and an assistant second.

2.4 Conclusion of Literature Review

The literature confirms that while the technology for voice control and automation exists, applying it specifically to *rural digital literacy* is a novel approach. We are building upon the technical foundations laid by these previous researchers but directing the purpose toward social impact.

METHODOLOGY

3.1 System Overview

Athena operates like a pipeline. It starts by listening to the user, converting that sound to text, understanding what the user wants (the "Intent"), and then routing that request to the correct tool—whether that's the computer's operating system or an external AI for answering questions.

3.2 System Architecture Diagram

(Note: Include a flowchart here like the one in your presentation slide "Algorithm Used". It should show: User Voice -> Mic Input -> Speech-to-Text API -> Intent Recognition -> Task Execution -> Output on Screen/Speaker)

3.3 Audio Capture & Speech-to-Text Module

This is the "ears" of our project. We use the **Py Audio** library to access the microphone. Once the audio is captured, it is sent to a **Speech-to-Text (STT) API** (like Google Speech Recognition or Deep Speech).

- **Why this API?** We chose these APIs because they are trained on massive datasets, meaning they can understand different accents and speaking speeds better than a simple offline model could.

3.4 Intent Recognition & Routing (The "Brain")

Once we have the text (e.g., "Please open the calculator"), the system needs to understand what to do.

- **JSON Dictionary:** We use a JSON-based "dictionary" where we map keywords to specific actions. For example, keywords like "teach," "open," or "create" trigger specific code functions.
- **Routing:** If the user asks a general question (e.g., "What is the capital of India?"), the router sends this to the **OpenAI API**. If the user gives a command (e.g., "Create a file"), it routes it to the OS Automation module.

3.5 Task Execution & OS Integration

This is where Athena actually does the work. We rely heavily on Python's native libraries:

- **OS & Subprocess Modules:** These allow Athena to talk directly to the Windows/Linux system to create folders, delete files, or launch software like Notepad.
- **Py Auto GUI:** For visual demonstration, this library allows Athena to take control of the mouse cursor and "show" the user where to click. This is crucial for the teaching aspect.

3.6 User Interface & Feedback

- **Kivy Framework:** We built the front-end using Kivy, a Python toolkit. It provides a clean, simple window that shows what Athena heard and what it is doing. This visual feedback reassures the user.
- **Eleven Labs API (TTS):** To make the assistant sound human and encouraging, we use Eleven Labs for Text-to-Speech. It sounds much more natural than robotic default voices, which helps beginners feel comfortable.

RESULTS AND DISCUSSION

4.1 Execution Environment

The project was developed and tested in a Python environment. We utilized **Visual Studio Code** for writing the logic and integrated various API keys (Google, OpenAI, Eleven Labs) for the different modules to talk to each other.

4.2 Interactive Analysis Interface

When the application launches, the Kivy interface appears, greeting the user with "Hello User, I am Athena". The interface is designed to be minimal text and high visual contrast for simple reading.

- **Test Case 1 (File Creation):** When we gave the command "Create a file," Athena successfully prompted for a name, found the desktop path, and created the text file. The user could see the file appear in real-time.
- **Test Case 2 (General Knowledge):** When asked "Who is the Prime Minister?", the query was successfully sent to the OpenAI API, and the answer was spoken back clearly.

4.3 Task Completion and Accuracy

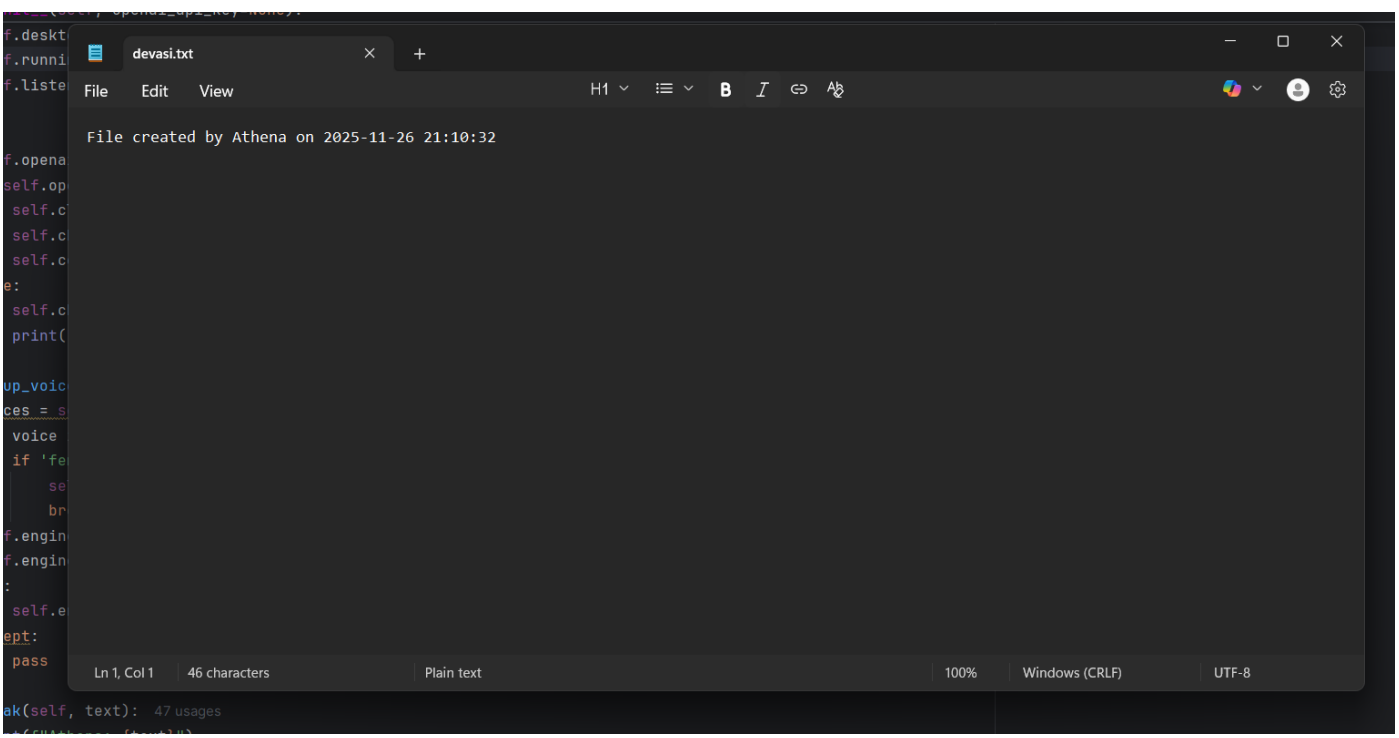
During our testing, the voice recognition worked accurately for clear English commands. The "Intent Router" successfully distinguished between a request to *do* something (OS command) and a request to *know* something (Generative AI query). The fail-safe measures prevented the system from crashing if a user asked for a file that didn't exist, instead providing a polite error message.

- Intro

```
C:\Users\Lenovo\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\Lenovo\PycharmProjects\PythonProject\athena.py
==================================================
ATHENA - AI Adaptive Assistant with ChatGPT
==================================================

Initializing Athena...

Commands you can use:
- 'search [query]' or 'google [query]'
- 'youtube [query]' or 'search youtube [query]'
- 'open [website]' (gmail, github, reddit, etc.)
- 'what time is it' or 'what's the date'
- 'create file [filename]'
- 'delete file [filename]'
- 'list files' or 'show files on desktop'
- 'open [app name]' (paint, calculator, notepad, etc.)
- 'remind me to [task]'
- 'shut down' or 'restart'
- 'clear conversation' - Clear ChatGPT history
- 'help' for full assistance
- 'exit' or 'goodbye' to quit
- Or just ask me any question! I'll use ChatGPT to answer!

==================================================

Athena: Hiii! I'm Athena, your personal AI assistant , I'm so excited to help you today! What can I do for you?
Pitch adjustment not supported when using SAPI5
Listening...
```

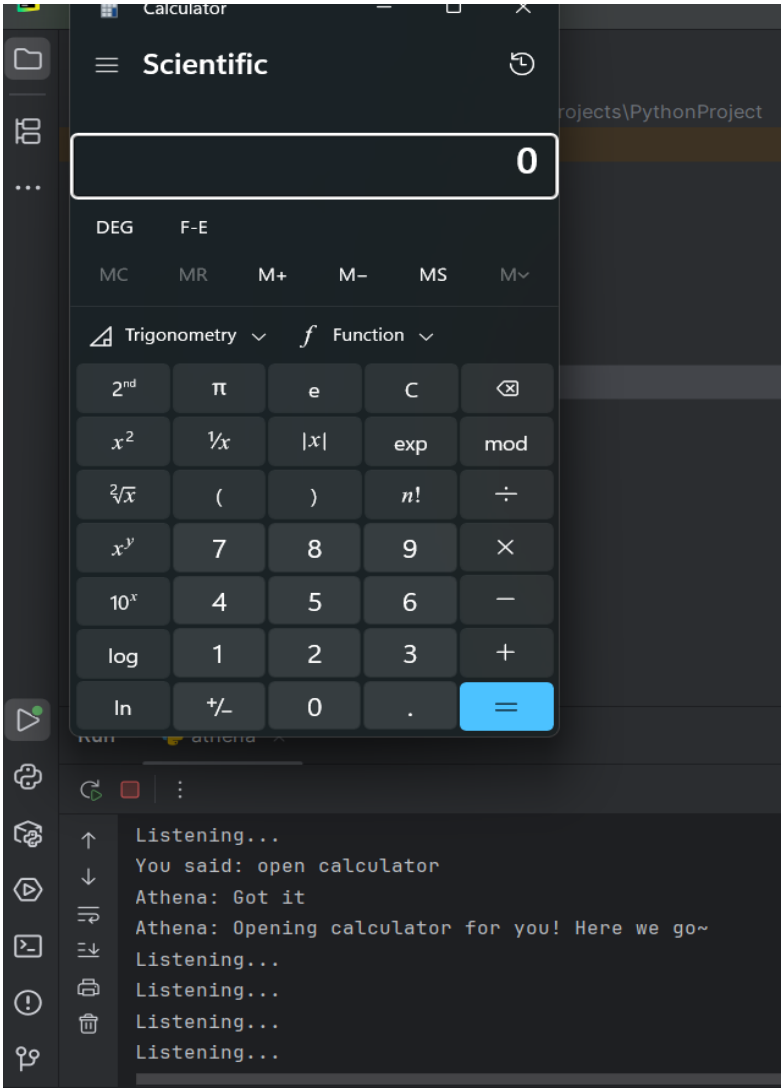
- Creating an text file

```
You said: create file named Devasi
Athena: Got it
Athena: Yay! Created devasi.txt on your desktop~
Listening...
```



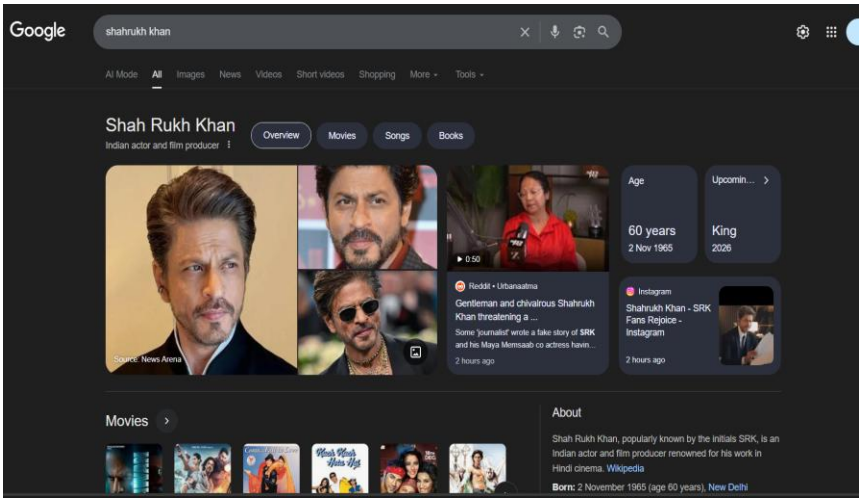
- Opening any app or software in pc

```
You said: open calculator
Athena: Got it
Athena: Opening calculator for you! Here we go~
Listening...
```



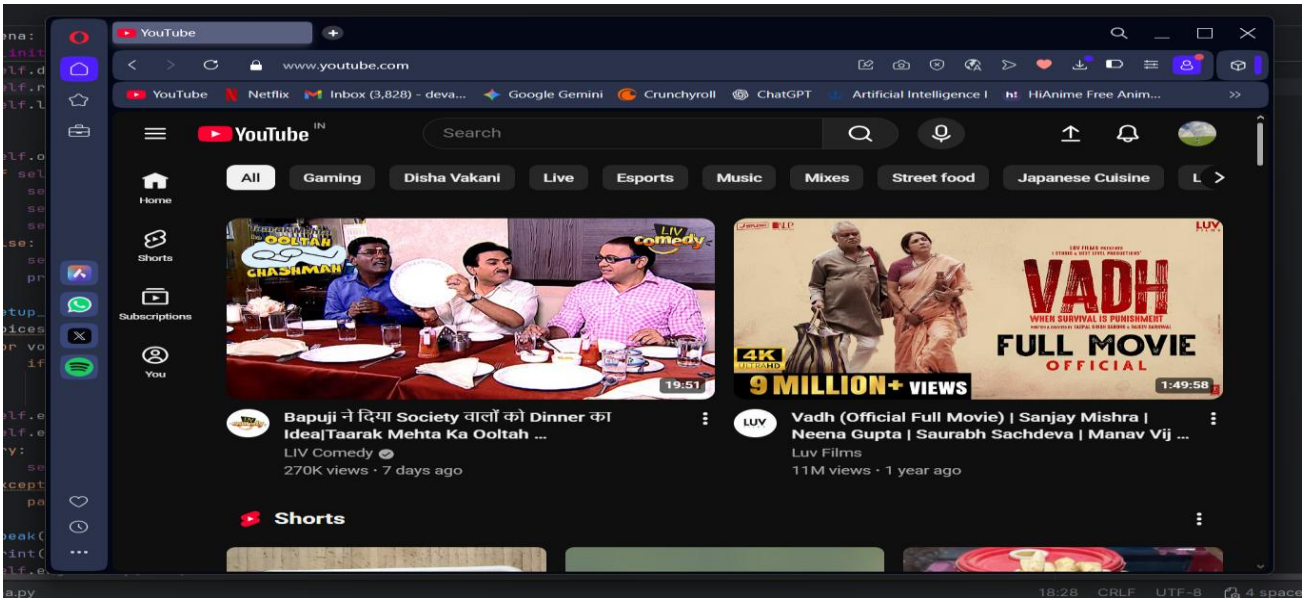
- Google search

```
You said: search Shahrukh Khan
Athena: Got it
Athena: Searching Google for shahrukh khan! Here you go~
Listening...
```



- Open any URL

```
You said: open YouTube
Athena: Got it
Athena: Opening youtube for you! Here we go~
Listening...
```



- Ask Date and Time

```
You said: what time it is
Athena: Got it
Athena: It's 09:09 PM right now!
Listening...
Listening...
```

- Èxit

```
Listening...
Listening...
You said: exit
Athena: Got it
Athena: Aww, goodbye! Have an amazing day! See you soon~
```

CONCLUSION & FUTURE SCOPE

5.1 Conclusion

Athena proves that AI can be more than just a productivity tool for experts; it can be a teacher for beginners. By combining voice recognition, OS automation, and a simple interface, we have created a system that empowers rural users to overcome their fear of computers. It moves beyond simple command execution to provide step-by-step guidance, effectively bridging the digital literacy gap.

5.2 Future Scope

To make Athena truly ready for mass adoption in rural India, we plan the following upgrades:

1. **Gamified Learning:** Adding points and levels to make learning fun for children.
2. **Full Local Language Support:** Integrating better translation APIs so users can speak entirely in Hindi, Marathi, or Tamil, and Athena will still understand.
3. **Government Scheme Integration:** Connecting Athena to government portals so it can help villagers apply for digital services and schemes automatically.

5.3 Individual Contribution

1. **Rahul Sharma:** Content Creation & Reseach**.**
2. **Arnav Anand:** Technical Skills.
3. **Devarshi Prabhakar:** Technical Skills.
4. **Aryan Kumar:** PPT Content Writing.
5. **Chirag Gartan:** PPT Content Writing.

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